

Experiment – 9

SELECT with various clauses – GROUP BY, HAVING, ORDER BY

```
mysql> INSERT INTO student (StudentID, Name, Department, Marks) VALUES
-> (101, 'Rahul', 'CSE', 85),
-> (102, 'Anjali', 'CSE', 90),
-> (103, 'Kiran', 'ECE', 75),
-> (104, 'Priya', 'ECE', 80),
-> (105, 'Arun', 'IT', 95),
-> (106, 'Sneha', 'IT', 88);
Query OK, 6 rows affected (0.06 sec)
Records: 6 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM student;
+-----+-----+-----+-----+
| StudentID | Name | Department | Marks |
+-----+-----+-----+-----+
| 101 | Rahul | CSE | 85 |
| 102 | Anjali | CSE | 90 |
| 103 | Kiran | ECE | 75 |
| 104 | Priya | ECE | 80 |
| 105 | Arun | IT | 95 |
| 106 | Sneha | IT | 88 |
+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> SELECT Department, AVG(Marks) AS AverageMarks
-> FROM student
-> GROUP BY Department;
+-----+-----+
| Department | AverageMarks |
+-----+-----+
| CSE | 87.5000 |
| ECE | 77.5000 |
| IT | 91.5000 |
+-----+-----+
3 rows in set (0.05 sec)
```

```
mysql> SELECT Department, AVG(Marks) AS AverageMarks
-> FROM student
-> GROUP BY Department
-> HAVING AVG(Marks) > 80;
+-----+-----+
| Department | AverageMarks |
+-----+-----+
| CSE | 87.5000 |
| IT | 91.5000 |
+-----+-----+
2 rows in set (0.05 sec)

mysql> SELECT *
-> FROM student
-> ORDER BY Marks DESC;
+-----+-----+-----+-----+
| StudentID | Name | Department | Marks |
+-----+-----+-----+-----+
| 105 | Arun | IT | 95 |
| 102 | Anjali | CSE | 90 |
| 106 | Sneha | IT | 88 |
| 101 | Rahul | CSE | 85 |
| 104 | Priya | ECE | 80 |
| 103 | Kiran | ECE | 75 |
+-----+-----+-----+-----+
6 rows in set (0.01 sec)

mysql> SELECT Department, AVG(Marks) AS AverageMarks
-> FROM student
-> GROUP BY Department
-> HAVING AVG(Marks) > 80
-> ORDER BY AverageMarks DESC;
+-----+-----+
| Department | AverageMarks |
+-----+-----+
| IT | 91.5000 |
| CSE | 87.5000 |
+-----+-----+
```